

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

303000094 - Geometría Diferencial Y Compleja Avanzadas

DEGREE PROGRAMME

30AH - Máster Universitario En Matemáticas Avanzadas

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 1

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1. Description

1.1. Subject details

Name of the subject	303000094 - Geometría Diferencial y Compleja Avanzadas
No of credits	6 ECTS
Type	Optional/elective
Academic year of the programme	First year
Semester of tuition	Semester 1
Tuition period	September-January
Tuition languages	English
Degree programme	30AH - Máster Universitario en Matemáticas Avanzadas
Centre	30 - Esc. Politéc. Enseñanza Superior (epes)
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Jesus Castro Infantes (Subject coordinator)		jesus.castro@upm.es	Sin horario. Contact by email
Juan Angel Rojo Carulli		juan.rojo.carulli@upm.es	Sin horario. Contact by email

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

The subject - recommended (passed), are not defined.

3.2. Other recommended learning outcomes

- Multivariable Calculus
- Complex Analysis
- Geometry of Curves and Surfaces

4. Skills and learning outcomes *

4.1. Skills to be learned

CO1 - Conocer los principales métodos y teorías matemáticas, próximos a la frontera del conocimiento, que permitan tener una amplia perspectiva en al menos tres de las siguientes áreas: Análisis Matemático, Métodos Numéricos, Ecuaciones Diferenciales, Estadística y Ciencia de Datos, Álgebra, Geometría y Topología. TIPO: Conocimientos o contenidos.

CO2 - Conocer demostraciones matemáticas avanzadas que permitan relacionar de una manera rigurosa las hipótesis con las conclusiones en al menos tres de las áreas relacionadas en el CO1. TIPO: Conocimientos o contenidos.

CO3 - Conocer diversas aplicaciones de las teorías matemáticas y su modelización en otros entornos o disciplinas que permitan adaptar los conocimientos matemáticos a nuevos problemas. TIPO: Conocimientos o contenidos.

CP2 - Ser capaz de trabajar en equipos multidisciplinares que involucren conocimientos matemáticos avanzados en áreas teóricas o aplicadas. TIPO: Competencias.

CP3 - Seleccionar y comprender la bibliografía científica sobre un tema concreto, extrayendo la información relevante. TIPO: Competencias.

CP4 - Comunicar de forma oral y escrita conocimientos y conclusiones científicas, valorando diferentes interpretaciones en públicos especializados y no especializados de una manera clara y rigurosa. TIPO: Competencias

CP5 - Complementar la formación matemática de manera autónoma en avances específicos de las Matemáticas, seleccionando la bibliografía adecuada y planteando cuestiones de interés. TIPO: Competencias

CP6 - Reflexionar sobre la relevancia social y ética de la aplicación de los conocimientos adquiridos a determinados problemas. TIPO: Competencias

HA1 - Aplicar los conocimientos matemáticos adquiridos a la resolución de problemas nuevos de una manera crítica, generando hipótesis de trabajo claras, definiciones precisas, nuevos enfoques con fundamento matemático y soluciones originales. TIPO: Habilidades o destrezas

HA2 - Ser capaz de abstraer y detectar las dificultades relevantes ante problemas complejos para poder plantearlos correctamente y resolverlos con técnicas matemáticas avanzadas. TIPO: Habilidades o destrezas

HA3 - Dominar el rigor matemático en las técnicas de demostración aprendidas para su posterior uso en el enunciado y demostraciones de nuevos teoremas. TIPO: Habilidades o destrezas

HA4 - Poseer destreza para proponer, validar e interpretar modelos matemáticos de diferente dificultad en situaciones reales complejas, utilizando las herramientas matemáticas más adecuadas a los fines que se persigan, y valorando las hipótesis y conclusiones asociadas a cada nivel de dificultad del modelo. TIPO: Habilidades o destrezas

HA5 - Adaptar teorías y técnicas matemáticas avanzadas al estudio de problemas en otros ámbitos con criterio científico para valorar su repercusión y formular las hipótesis adecuadas. TIPO: Habilidades o destrezas

HA6 - Interpretar adecuadamente la relevancia de nuevos resultados en Matemáticas, su aplicabilidad en situaciones particulares donde pudieran suponer nuevos avances y su relación con los resultados existentes. TIPO: Habilidades o destrezas

4.2. Learning outcomes

RA2 - Los resultados de aprendizaje en las tres categorías propuestas: conocimientos/contenidos, habilidades y competencias son los señalados en la memoria del título

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

This course aims to cover some of the most important aspects within the fields of Riemannian geometry and complex geometry, with a particular emphasis on the strong interconnections between the two areas. To this end, the course content will be framed within the general theoretical framework and the study of specific examples that help to reinforce the theory and build intuition. The goal is to strike a balance between generality (theoretical concepts in arbitrary dimension) and concreteness (specific examples and results in low dimensions).

5.2. Syllabus

1. Differential and Riemannian Geometry

- 1.1. Differentiable manifolds. Riemannian metrics.
- 1.2. Calculus on Riemannian manifolds. Levi-Civita connection. Gradient, Hessian, and Laplacian of a function. Geodesics and the exponential map. Hopf-Rinow theorem.
- 1.3. Curvatures on a Riemannian manifold. Spaces of constant sectional curvature.
- 1.4. Isometric immersions. Second fundamental form. Fundamental equations.

2. Complex Geometry

- 2.1. Complex manifolds. Complex structures on a differentiable manifold.
- 2.2. Projective space, affine and projective manifolds.
- 2.3. Hermitian and Kähler metrics. Some properties of Kähler manifolds.
- 2.4. Complex curves and Riemann surfaces. Holomorphic and meromorphic functions, and branched coverings.
- 2.5. Embedding of projective curves.
- 2.6. Plane curves. Genus-degree formula.
- 2.7. Complex tori and elliptic curves. Group structure and applications in cryptography.

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	Lecture and practical session Duration: 04:00 Lecture			
2	Lecture and practical session Duration: 04:00 Lecture			
3	Lecture and practical session Duration: 04:00 Lecture			
4	Lecture and practical session Duration: 04:00 Lecture			
5	Lecture and practical session Duration: 04:00 Lecture			
6	Lecture and practical session Duration: 04:00 Lecture			Submission of solved problems Individual work Progressive assessment Not Presential Duration: 00:00
7	Lecture and practical session Duration: 04:00 Lecture			
8	Lecture and practical session Duration: 04:00 Lecture			
9	Lecture and practical session Duration: 04:00 Lecture			
10	Lecture and practical session Duration: 04:00 Lecture			
11	Lecture and practical session Duration: 04:00 Lecture			
12	Lecture and practical session Duration: 04:00 Lecture			
13	Lecture and practical session Duration: 04:00 Lecture			Submission of solved problems Individual work Progressive assessment Not Presential Duration: 00:00

14	Lecture and practical session Duration: 04:00 Lecture			
15	Lecture and practical session Duration: 04:00 Lecture			Oral presentations Group presentation in the classroom Progressive assessment Presential Duration: 02:00
16				
17				Final exam Written test Progressive assessment Presential Duration: 02:00 Comprehensive Final Exam Written test Global examination Presential Duration: 03:00

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
6	Submission of solved problems	Individual work	No Presential	00:00	15%	/ 10	
13	Submission of solved problems	Individual work	No Presential	00:00	15%	/ 10	CP3 CP5 CO3 HA3 CP2 CP6 CO2 CO1 HA1 HA2 HA4 HA5 HA6
15	Oral presentations	Group presentation in the classroom	Face-to-face	02:00	10%	/ 10	CP3 CP4 CP5 CO3 CP2 CP6 CO2 CO1 HA1 HA2 HA4 HA5 HA6 HA3
17	Final exam	Written test	Face-to-face	02:00	60%	/ 10	CP3 CP4 CP5 HA1 CO3 HA3 CP6 CO2 CO1 HA2 HA4

							HA5
							HA6

7.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
17	Comprehensive Final Exam	Written test	Face-to-face	03:00	100%	5 / 10	CO3 HA3 CP6 CO2 CO1 HA1 HA2 CP3 CP5 HA4 HA5 HA6

7.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Retake exam	Written test	Face-to-face	03:00	100%	5 / 10	CP3 CP4 CP5 CO3 HA3 CP6 CO2 CO1 HA1 HA2 HA4 HA5 HA6

7.2. Assessment criteria

Progressive assessment :

15% First Submission of solved problems.

15% Second Submission of solved problems.

10% Oral presentation of group project.

60% Final exam including theoretical contents and exercises.

Global assessment:

100% Final exam including theoretical contents and exercises.

Retake exam

100% Final exam including theoretical contents and exercises.

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
M. P. Do Carmo: Riemannian geometry. Mathematics: Theory & Applications. Birkhäuser Boston, Inc., Boston, MA, 1992.	Bibliography	
M. SPIVAK, A comprehensive introduction to Differential Geometry I-V, Publish or Perish (1979).	Bibliography	

V. Muñoz, A. González-Prieto, J. Rojo, Geometry and Topology of Manifolds: surfaces and beyond, American Mathematical Society, 2020.	Bibliography	
F. Kirwan, Complex Algebraic Curves, London Mathematical Society, 1992.	Bibliography	
Huybrechts, Complex Geometry, Springer.	Bibliography	